December 15, 1960

Memorandum to: Dr. E. R. Piore

Subject: Farmboy

There have been some further developments here that you may want to know about.

1. Farmboy (also Stretch programming) has been transferred from Dunwell to Harmon. No internal changes are contemplated that I know of. I do not think this move will have much direct effect on the project, but it makes sense organizationally, would facilitate emergency measures if the need arises, and makes a fresh start possible in some of our dealings with the Agency.

2. Several weeks ago the Agency sounded out CEIR and Computer Usage Corporation on whether one of them could do the Alpha processor instead of IBM. CEIR is said to have answered that they could do the job in a year and at one-third the price IBM is charging. CUC is said to have expressed "great interest".

3. This coming Monday, December 19, Harmon and Campbell from DSD and Green and Oldfield from FSD will meet with Agency representatives to discuss the whole Farmboy matter. Today, December 15, Walter Jacobs from the Agency is visiting Poughkeepsie, and doubtless his reaction to the situation will be sought in advance.

4. In anticipation of these meetings, there have recently been a number of discussions between various people from FSD (Horton, Oldfield, Green and Kolsky) and DSD (DeCarlo, Dunwell, Harmon, Heising, Campbell, Herwitz, and Gerberich), some of which I have attended. The main points which have emerged and which will be the basis of the discussion with the Agency are (in my opinion) the following:

a. The schedule laid down for Alpha in September is being met so far. There is some slippage in the central planning activity, which is disturbing, but even there a good amount has been accomplished, and some of the other work is ahead of schedule. Thus, we continue to believe we can deliver the full Alpha system (field test version) by April, 1962. A truncated system could be ready by January 1962, and a severely truncated system (in fact, Autocoder only), by October 1961.

The cost of the entire Farmboy effort under these three alternatives would be 1.9, 1.7, and 1.5 million dollars, respectively. IBM will offer these three alternatives to the Agency, and will recommend the first.

b. DSD maintained (and I agree, though FSD was only half convinced) that these times and costs cannot be improved. In particular, there was further discussion of the central technical decision which DSD has taken, namely to structure the Alpha processor in such a way that a large part of it will be usable for Fortran, Cobol, etc., and especially whether this decision is costing the Agency either time or money. One is bound to have some misgivings on this matter, because the technical problem is made considerably more complicated by this approach, but I believe the advantages outweigh the disadvantages and that the difficulties are being solved fast enough, and the FSD people were sufficiently convinced not to register any objection.

c. It was also maintained and generally admitted that these times and costs are not bad from the Agency's point of view. They can have the full Alpha system within six months of what now appears the probable machine delivery date, and the really indispensable programming packages, MCP and Hasti, will be ready with the machine. For the full Alpha system they will be paying, from this time forward, about $700,000. It is assumed that CEIR's offer for Alpha meant one-third of IBM's total of 1.9 million; i.e. $633,000.

d. Furthermore, it was maintained and admitted that no competitor can beat these times, because of having to start from scratch, clear a team of programmers, and the rate at which machine time for debugging will become available precludes any much faster schedule. CEIR's statement that they can do the full Alpha in a year is regarded as impossible. As to cost, it was suggested that CEIR would be willing to take a loss for the sake of the contact with the Agency.

e. Apparently the Agency feels that IBM's plan requires them to subsidise an Autocoder which will be used more by others than themselves. The actual figures are that of the $700,000, which they would pay, about $180,000 will be their share of basic systems subroutines (Stretch is paying the other $180,000 for these routines); the remaining $520,000 is all for specifically Harvest programming.

f. Finally, there was general feeling, shared by DSD, that meeting the proposed schedule will not be easy. Machine time is one problem. The plan is to give the space now occupied by the AEC Stretch machine to the Farmboy group when that machine goes; this will put the programmers right next to the Harvest machine so that they can get full advantage of any available time on it. This is a good plan for the period while the

machine is in Poughkeepsie. But the more important period is after the machine goes to Washington, and arrangements for IBM personnel to use the machine there are not yet satisfactory. This is a matter of the greatest importance.

Also, FSD felt that the proposed schedule does not spell out enough objective milestones to make progress, or its absence, easily measurable. The discussion did provide two such milestones -- an Alpha manual and an Autocoder manual -- to be reached in the next 2-3 months.

5. I think these discussion did a great deal to pull FSD and DSD together over Farmboy. For this reason, and because Farmboy will continue to need close watching, I think that a monthly IBM review preceding the monthly meeting with the Agency would be worthwhile.

David Sayre

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